When Is the Edit Decided? A Layer–Time Lens on How a Diffusion Editor Thinks with an Internal Image

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Abstract

Between reading "turn the car red" and painting it, what does a diffusion image editor do? The mechanics suggest gradual development – sixty transformer blocks times twenty denoising steps, the edit surfacing progressively like a photograph in developer fluid. We show this picture is wrong in a measurable way. We build the first tuned lens for a diffusion-transformer (DiT) image editor, decoding intermediate residual streams through the model's own frozen output head, and pair every passive reading with causal interventions – ablation, transplant, and steering. The editor thinks with an internal image: the edit is *decided early* – linearly readable by layer 6, where the frozen head still reads noise – and only *translated late*, in a sharp, timestep-invariant band (layers 52–54 of 60, transition width as narrow as one layer) from an internal target-image code into the flow field that paints it. This translation boundary is edit-family universal and survives an independently fine-tuned checkpoint and a four-step distilled model. Readability, causal necessity, and linear writability dissociate three distinct ways, and we map exactly where. Because the readings are causal, they convert into training-free levers – including truncating classifier-free guidance after step two for a 45% compute saving with all edit families visually intact.

Introduction

Between reading "turn the car red" and painting it, what does a diffusion image editor actually do? The mechanics of diffusion suggest a gradual-development story, each of sixty blocks and twenty denoising steps contributing its small share – but this paper measures that story and finds it wrong. In the model we study, an edit is not developed but *decided early* – linearly present within the first handful of blocks – and only *translated late*, in the last few blocks, from an internal goal-image code into the flow field that does the painting.

Watching this requires an instrument that did not exist. The logit lens (nostalgebraist 2020) and its learned successor, the tuned lens (Belrose et al. 2023), let us watch a language model's next-token prediction take shape layer by layer by projecting each intermediate residual stream through the model's own final unembedding. Diffusion transformers (DiTs) that edit images (Peebles and Xie 2023; Qwen Team 2025) have no analogous tool: there is no vocabulary for an image patch's hidden state, so "what is this layer currently drawing" has been answerable only by running the whole model to the end. We build a layer–time lens for a production DiT editor, Qwen-Image-Edit-2511 (60 blocks), by decoding intermediate residual streams through the model's own frozen output head, and – crucially – pair every passive reading with linear probes and *causal* interventions (ablation, transplant, steering) that test whether the activations the lens points to are the ones that actually decide the edit.

Naively porting the logit-lens recipe reproduces the familiar mid-stack "nothing happens yet" appearance – and that appearance is itself misleading. A linear probe trained on the very activations the frozen head cannot decode reaches near-perfect cross-seed accuracy 40+ layers earlier (0.88 at layer 6, where the head reads 0.005–0.14). The frozen head is reading the network's *output-space* code; the decision is present far earlier, just held in a different basis until a narrow band near the top of the stack translates it. This places the work in the "thinking with images" agenda (Su, Xia et al. 2025), but at the mechanistic rather than the interface level: the intermediate image other work hands a model explicitly, we find already present *latently* inside a generative editor – formed early, readable, steerable, transplantable, and turned to pixels only at the end.

The main contributions of this work are as follows:

- **An instrument.** The first training-free, frozen-head layer–time, tuned, *and* differential lens for a DiT image editor, paired with linear probes and three intervention primitives (ablate, transplant, steer) with stated write semantics, under a falsification-first protocol – pre-registered predictions, per-image visual verdicts, and bit-exact gates.
- **A mechanism: the editor thinks with an internal image.** The edit is decided early (linearly readable by layer 6), carried as a target-image code for most of the depth, and translated into the velocity code only in a sharp, timestep-invariant band (layers 52–54, transition width one layer) that is edit-family universal and survives an independent fine-tune and a 4-step distillation. Readability, causal necessity, and linear writability dissociate three different ways, and we map where.
- **Levers.** Because the readings are causal they become training-free tools – most cleanly, truncating classifier-free guidance after step 2 saves 45% of forward passes with all edit families intact, isolating CFG's real job as scope resolution rather than deciding the edit.

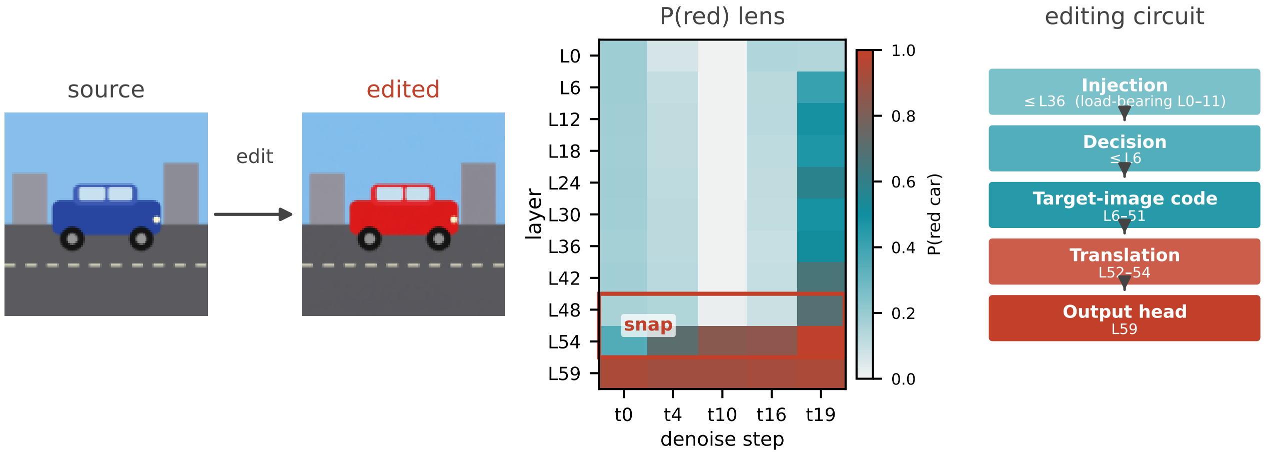

Figure 1: A layer–time lens for a DiT image editor. Decoding the frozen output head at intermediate layers of *car_red* gives a P(red car) grid across layer and denoising step (center); a sharp snap between layers 48 and 54 is where the head-readable signal first appears, but a linear probe shows the edit is already linearly decodable from layer 6. We read this as a five-stage *editing circuit* (right): the instruction is injected early, the edit decision is linearly readable by layer 6, most of the stack then carries a target-image code, a narrow band (L52–54) translates it into the velocity code, and layer 59's frozen head reads it out – the edit is decided early and translated late, not developed gradually.

Every intervention was pre-registered and every verdict made on the images themselves; four of our own initial misreadings are retained as data. This is exploratory, single-model-family work, and we are explicit about where the lens is unreliable.

Related Work

Logit and tuned lenses. The logit lens (nostalgebraist 2020) projects intermediate residual streams through a language model's frozen unembedding; the tuned lens (Belrose et al. 2023) fits a small per-layer affine map instead. Both were built for autoregressive text models, where the unembedding gives every layer a shared, well-defined target vocabulary; a DiT editor has no such vocabulary; a patch's hidden state is not a distribution over anything until the whole network has run. We port both lens families to this setting – staying on the frozen-head side for honesty and crossing to the learned side only to render the internal image – with the linear representation hypothesis (Park, Choe, and Veitch 2024) licensing the informative disagreement between a frozen head and a probe: if the network represents "red" along a roughly linear direction, a probe can find it long before the frozen head's fixed final-layer projection can.

Instruction-based diffusion image editing. Two lines established how a diffusion model can be made to follow an edit instruction at all. InstructPix2Pix (Brooks, Holynski, and Efros 2023) trains directly on synthesized instruction-image pairs so the model learns the edit end to end; Prompt-to-Prompt (Hertz et al. 2023) instead manipulates the model's cross-attention maps at inference time to inject the edit while preserving the source's structure, training-free. Both contribute a *mechanism for producing* the edit; neither asks what representation the network forms once an instruction is given, or where in the forward pass that representation becomes the pixels that change. Qwen-Image-Edit-2511 is exactly such an instruction-following editor, and our lens targets that open question directly – not a new editing mechanism, but a reading of the one a production model already has.

Thinking with images, and diffusion interpretability. A growing line has multimodal models reason *with* explicit intermediate images – sketching, annotating, or otherwise externalizing a visual scratchpad as part of the reasoning trace (Su, Xia et al. 2025; Hu et al. 2024; Wu et al. 2024; Menon, Zemel, and Vondrick 2024; Li et al. 2025), with production systems now shipping the idea too (OpenAI 2025). We supply the mechanistic complement: an image that exists *latently* inside a generative editor before any pixel is committed, with no explicit scratchpad or tool call involved. The Diffusion Lens (Toker et al. 2024) decodes a text-to-image model's *text encoder*; we instead decode the editing DiT's own denoising stack mid-generation, building on the flow-matching and rectified-flow formulation (Lipman et al. 2023; Liu, Gong, and Liu 2023) and classifier-free guidance (Ho and Salimans 2022) that Qwen-Image-Edit-2511 (Qwen Team 2025; Peebles and Xie 2023) uses.

Causal interpretability. Causal tracing / ROME (Meng et al. 2022) patches activations between runs, the move behind our transplant; Heimersheim and Nanda (2024) catalogue the patching pitfalls that motivate reporting "readable" and "necessary" separately, and activation addition (Turner et al. 2023) is the steering primitive we instantiate with a probe direction. All three were developed and validated on language models with a fixed token vocabulary at every patched site; we port the same three primitives (ablate, transplant, steer) to a continuous, spatial image stream, where a "patch" is a region of pixels rather than a token and sufficiency must be checked against a rendered image rather than a next-token distribution. Our semantic readout uses CLIP (Radford et al. 2021) for lack of a better zero-shot alternative.

Proposed Method: A Layer–Time Lens and Causal Probes

We study Qwen-Image-Edit-2511, a 60-block DiT editor whose image stream concatenates noisy target tokens with always-clean condition tokens (Figure 2). All quantities below are read from the *post-block residual* of selected blocks via forward hooks, captured in the model's native `bfloat16`: image-stream activations reach absmax $\sim 3\times 10^9$ even at layer 0, so a cast to `float16` silently overflows most elements to `inf` (bf16 capture is a lossless copy, not a downcast).

Four lenses. The *time lens* decodes the flow-matching clean-image estimate $\hat{x}_0 = x_t - \sigma_t v_\theta(x_t, t)$ through the VAE at step t: what the model currently believes it will output. The *depth lens* asks the analogous question across *depth*: it feeds an intermediate layer's hidden state through the model's own frozen final head (a non-affine `AdaLayerNormContinuous` + linear `proj_out`, bit-exact identical whether invoked at layer 59 or re-invoked earlier) and decodes the resulting pseudo-velocity to an $\hat{x}_0$ preview. Because that head's LayerNorm is non-affine, rescaling an intermediate state to the terminal norm is a no-op in principle and in practice (raw vs. rescaled agree to `max_rel_diff`$=0$ on every grid; verified three ways) – the *raw* head is the honest lens, and a standing self-consistency gate accompanies every grid we report. The *tuned lens* (Belrose et al. 2023) crosses to the learned side for one purpose – rendering the mid-stack code back into an *image* at every depth: a per-(layer, timestep) ridge regression from the hidden state (3072-d) to the conditional-pass velocity (64-d), one dictionary per cell, with λ chosen on a held-out token split.

A differential lens. The tuned lens renders the whole internal image; to isolate the *edit* from the reconstruction, we additionally difference it against a control. We run the same source and seed twice – once with the edit prompt, once with the literal instruction "keep the image unchanged" – and decode both through the same frozen dictionaries (no refit). Because the two runs share a seed, their step-0 latents are bit-identical, so $D_{\ell,t} = |\mathrm{decode}(h_\ell^{\mathrm{edit}}) - \mathrm{decode}(h_\ell^{\mathrm{unchanged}})|$ isolates the internal edit plan, the reconstruction cancelling by construction. A control-fidelity gate – the correlation between the no-op run and the untouched source – flags the rare case where the model cannot leave an image alone before any differential number is trusted; because the control is independent of the edit prompt, it can be computed once per source image and cached across many candidate edits.

Semantic readout. Neither lens is itself a score; each produces an image. We crop the edited region and score it zero-shot against a small bank of natural-language labels with CLIP (Radford et al. 2021), reporting the target-label probability. This is a whole-crop judgment, not a per-patch one, and inherits CLIP's failure modes on out-of-distribution previews; we gate on decode coherence and flag every violation.

Causal probes. Reading a lens shows the representation is *consistent with* a hypothesis, not that it is *causal*. Three interventions, all forward-hook overwrites of the post-block residual, test causality. **Ablate**: zero or mean-replace selected target rows at selected layers (an empty spec reproduces the baseline bit-for-bit). **Transplant**: overwrite a recipient run's rows with a donor run's, matched by layer/step/patch – testing *sufficiency*. **Steer**: add a probe direction $\alpha\, d$ (in units of per-row RMS) without a donor pass – testing *linear writability*. All three are applied identically to both CFG passes to isolate the intervention rather than a lopsided guidance combination.

Protocol: predict, look, gate. Three rules make the *reasoning* reproducible, not just the runs. **Predict, then run**: every intervention was pre-registered; refutations are reported as prominently as confirmations, and several headline results are pre-registered predictions that failed informatively. **Look at every image**: no verdict rests on CLIP alone – every run carries a per-image visual verdict recorded before the aggregates were read, and where the two disagree the image wins. **Gate the instrument**: the empty-ablation and raw-vs-rescaled checks are standing regressions re-run per experiment. Porting the method to another editor needs only a native-dtype residual hook, the frozen-head gate, and region-restricted probes – exactly the recipe executed unchanged in the cross-model replication reported below under Generalization Across Checkpoints.

Experiments

Experimental Setup

We evaluate on Qwen-Image-Edit-2511, a 60-block DiT editor, using a battery of edit instructions spanning five categories – color, background, style, removal, and addition/insertion – run on natural photographs and one flat-art source. Battery size varies with what is being measured (10 cases across the five categories for the time-lens commit-step sweep and for the differential-preview lever, 20 for the finer IoU-based onset measurement and for the CFG-truncation check), but the same five families recur throughout, and every quantitative number is paired with a per-image visual verdict recorded before the aggregate is read, following the predict-look-gate protocol described above. Three readouts do the reporting: the frozen output head's decode, scored zero-shot with CLIP against a small label bank; a cross-seed linear probe, fit on one seed's hidden states and evaluated on a held-out seed; and, for the tuned lens, held-out R^2 against the true conditional-pass velocity. Two further checkpoints sharing Qwen-Image-Edit-2511's transformer classes – FireRed-Image-Edit-1.1, an independent fine-tune, and Rapid-AIO, a 4-step distilled merge – let us ask whether a finding belongs to the architecture or to one training run, tested directly below under Generalization Across Checkpoints.

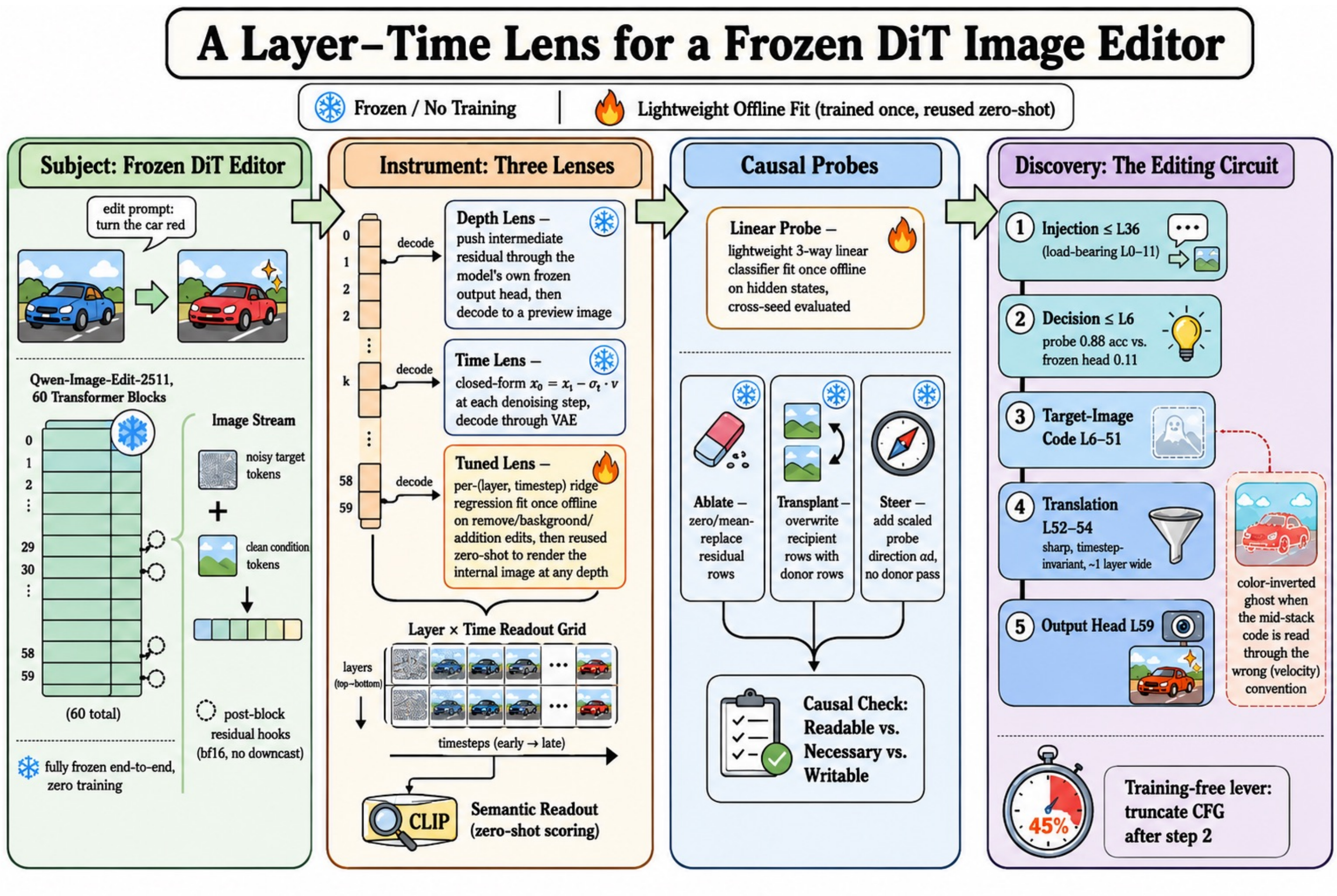

Figure 2: Pipeline. The image stream concatenates noisy target tokens with always-clean condition tokens; each block's post-block residual is tapped and read two ways – the *depth lens* pushes it through the model's own frozen output head before decoding, the *time lens* instead reads the current step's head output combined with the current latent ($\hat{x}_0 = x_t - \sigma_t v$). Both routes end in a decode-then-CLIP readout, computed identically for both true-CFG passes.

The Edit Is Decided Early

In time. If the edit surfaced progressively, the commit step would be late and roughly uniform. Running the time lens over a battery of 10 cases across 5 edit categories, the step at which $P(\text{target})$ first reaches within 0.05 of its final value is instead *edit-dependent* and mostly *immediate*: color, background, and style edits commit at step 0–2, and only a wholesale new-object insertion (*add_boat*) commits as late as step 7. A finer IoU-mask-based onset measurement over 20 cases shows the split is categorical, not continuous: 18 of 20 – spanning final-edit-area fraction 0.003 to 0.99 – onset at step 0 regardless of how large or novel-looking the finished edit is, and no continuous proxy predicts the delay (edge-novelty $\rho=0.24$, $p=0.30$; area $\rho=-0.44$, running the *wrong* way). Only the two edits that insert genuinely novel structure with no existing pixels to build on – *add_boat* and *add_runner* – start late.

In depth, the naive lens says "late". The frozen depth lens tells a different-looking story. Decoding *car_red*'s intermediate layers through the model's own head, the edit becomes head-readable only in a sharp snap between layers 48 and 54 of 60 – exactly the familiar mid-stack "nothing happens yet" appearance of an LLM logit lens.

But a probe reads it 40 layers earlier. That appearance is misleading. A 3-way cross-seed linear probe (which of three colors was requested), fit on hidden states from one seed and evaluated on a held-out seed, is already at 0.883 accuracy at **layer 6**, step 4 – five blocks into a 60-block network – where the frozen head at the identical cell reads only $P(\text{red car})=0.111$ (Figure 3). The earliest cell to cross 0.9 is L36/step 0. The decision is present far earlier than the head can read it; it is simply held in a basis the head – built for the network's final-layer convention – cannot parse until the last few blocks. Decided early, read out late.

Where the instruction enters. A complementary text-stream ablation locates where the prompt is *consumed*: zeroing the model's text tokens in only the first 11 blocks garbles the whole scene, while zeroing them after layer 36 barely dents the edit ($P(\text{red})=0.993$). The instruction is fused into the image stream early and is functionally complete by layer 36 – the "injection" stage of the circuit in Figure 1, whose full

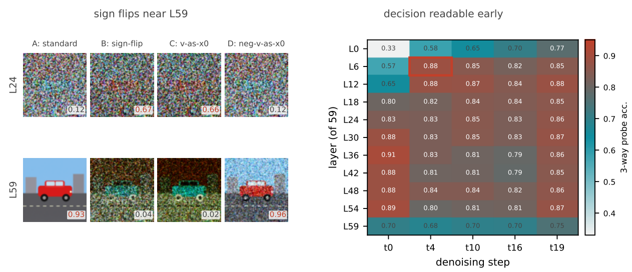

Figure 3: Left: four ways of turning an intermediate layer's frozen-head output into an image for *car_red* at $t{=}4$ (A: standard $\hat{x}_0{=}x_t{-}\sigma_t v_\ell$; B: sign-flipped; C: v_ℓ decoded directly; D: $-v_\ell$ decoded directly) – at L24 the standard decode is wrong-colored while the sign-flipped/direct decodes already look red, and at L59 this reverses. Right: cross-seed 3-way probe accuracy (chance $=0.33$) by layer/step; the headline cell L6/t4 (0.883) is far above the frozen-head reading at the same cell, and probe accuracy drops at L59 relative to L48–L54.

text-band table is in the supplement. Together with the probe, this gives the paper's five-stage picture: injection ($\leq$L36), decision ($\leq$L6), a target-image code, translation (L52–54), and the output head.

The Edit Is Translated Late: Two Codes

If the decision is present by layer 6 but the frozen head cannot read it until layer 52, the mid-stack must hold the edit in a *different code*. Three measurements show what that code is and where it is converted.

Two decoding conventions cross over. At an intermediate layer the frozen head yields a per-token vector v_ℓ we normally treat as a velocity, forming $\hat{x}_0 = x_t - \sigma_t v_\ell$ (variant A). Decoding v_ℓ instead *directly*, as if it were already a clean latent (variant C), inverts the story: for *car_red* at step 4, at L24 the standard decode A scores $P(\text{red}){=}0.12$ (still wrong-colored) while the direct decode C already scores 0.66; at L59 this reverses (A $=0.93$, C $=0.02$). The mid-stack is not blank – it encodes the target content in an *image-like* convention, and a narrow band near the top reconciles it to the velocity convention the head consumes. (This is a whole-vector reweighting, not a literal sign flip: essentially no individual channel of v_ℓ has a negative slope against v_{59}, mean 0.003.)

The conversion is a sharp, timestep-invariant band. A dense every-layer sweep from L44 to L59 (Figure 4) locates the A/C crossover precisely: transition midpoint L53.6 at step 16 (width one layer) and L52.7 at step 4, the two differing by under one layer – the boundary's *location* is timestep-invariant. Crucially, the cross-seed probe accuracy is close to *flat* (0.77–0.84) across every layer from L44 through L58 and drops only at the terminal layer L59 (0.68–0.70). So the translation that changes what the head can read (L52–54) and the loss of linear separability (L59) are two *distinct* events sharing a neighborhood: the target-image code survives the translation intact and is degraded only by the final, velocity-specific projection.

A tuned lens renders the internal image, layer by layer. To confirm the mid-stack code is a genuine *image* rather than a probe artifact, we fit a per-(layer, timestep) tuned lens, as described above, on *remove*/*background*/*addition* edits and evaluate it *zero-shot* on the held-out *car_red* color family. Held-out R^2 rises with depth from 0.73–0.89 at L6 to 0.998 at L59, where a standing identity gate confirms the target is correct (decoding L59 through the frozen head reproduces the true velocity bit-for-bit, $\max|\text{diff}|{=}0$). Every depth down to L6 decodes a coherent scene: a pre-registered prediction that readability itself would have a boundary is refuted – the L52–54 boundary is a property of the frozen head's one fixed projection, not of the representation. At step 0 the pixels are pure noise, so the decode comes entirely from the velocity code, and we watch the edit enter the internal image directly (Figure 5): the source's blue car persists through L24, turns orange-red by L36, the scene composes by L44, and the car is fully red by L52. The same frozen dictionaries transfer with zero retraining to six further held-out cases (style, more colors, removal, background, addition); *add_runner* even reproduces the delayed onset on the internal image's own time axis – an empty path at step 0, a blurred figure by step 2, a clear runner by step 4.

The anti-image, mechanized. This also explains a striking artifact: the mid-stack *frozen*-head decode at later steps is a color-*inverted* ghost – a cyan car under a red-brown sky, the exact complement of the true scene. It is the image-like mid-stack code forced through a head built to consume a velocity: $\hat{x}_0 = x_t - \sigma v$ flips the sign of an image-like quantity. The tuned lens at the same cell is already coherent.

Readable, Necessary, Writable: Three Dissociations

Reading a lens shows only that a representation is *consistent with* an edit. Causal probes reveal that readability, causal necessity, and linear writability are three different properties, and a layer band can have any two without the third.

Readable $\neq$ necessary. Ablating *car_red*'s early layers (L0–11) is catastrophic ($P(\text{red}){=}0.049$) while ablating the late band L48–59 is nearly harmless (0.826, vs. an unablated baseline near 0.93) – even though that late band is exactly where the frozen head finally reads the edit. *beach_background* mirrors this (L0–11 ablation only partial at 0.193, L48–59 catastrophic at 0.040), and the split tracks commit time: the early-committing edit needs early layers, the late-committing edit needs late ones (Figure 6).

Sufficient $\neq$ necessary. Transplanting a *car_red* donor's car-region hidden states into a *car_green* recipient (which alone scores $P(\text{red}){=}0.011$) flips the output to red from *any* of four donor bands – including **L48–59** (0.940), the very band ablation just showed is dispensable to the recipient. The donor's late representation is *sufficient* to install the edit even where the recipient's own late computation is not *necessary*. A content control confirms specificity: splicing the same donor's *background*-region tokens (right band, wrong content) yields a garbled "blue car" (0.991), not red.

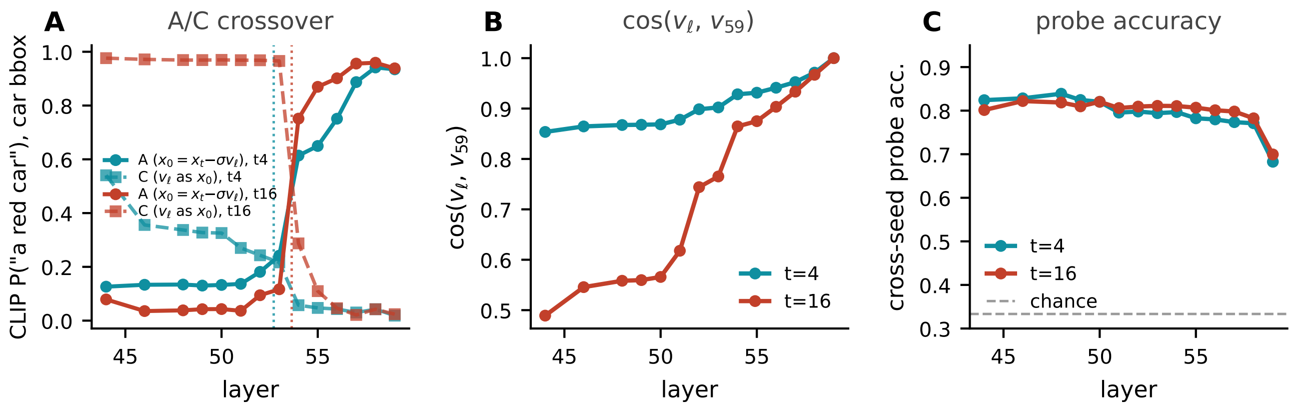

Figure 4: A dense, every-layer sweep of *car_red* over L44–L59 sharpens the coarse L48/L54 snap to a precise boundary. **A**: the standard decode and the direct-v decode cross between L52 and L54 at both steps shown. **B**: cosine similarity of the intermediate pseudo-velocity to the terminal v_{59} rises sharply in the same band. **C**: cross-seed probe accuracy is flat from L44 through L58 and drops only at the terminal layer.

Writable, but only onto a carrier. A single stored *direction* – the probe's d_{red}, no donor pass – added to the recipient's car rows gives a sharp sigmoidal dose-response (0.024 at $\alpha{=}0.5$, 0.977 at $\alpha{=}1$, saturating to $\alpha{=}8$); a random direction of equal norm is inert (0.017), and the negated direction overshoots to blue (0.945), reproducing the anti-image under an active intervention. But writability is bounded: at effective doses the steered region is not a recolored car but a flat red patch that *replaces* the car's geometry – the direction installs *which* hue, not *where to stop*. Supplying that missing "where" with the edited object's own silhouette mask turns every previously-failing dose into a clean in-place recolor. And conjuring an *object* outruns any single direction entirely: the probe direction for object identity is inert, because the read and write subspaces differ – a rank ladder over the donor−recipient difference resolves identity coarse-to-fine (category at rank 1, a coherent instance by rank 4, donor-faithful by rank 64, 2% of the model dimension). Readability, necessity, and writability thus dissociate three distinct ways, and we map exactly where.

A Preview Lever: The Differential Lens at 15% NFE

The differential lens defined in *Proposed Method* isolates the internal edit plan from the reconstruction; before using it as a lever we first check it is not simply picking up easy background variance rather than the edit itself. This also answers the open caveat raised above, in *The Edit Is Translated Late*, that the tuned lens's high R^2 might be incidental scene detail: token-level R^2 *inside* the edit region is 0.88–0.97, matching the full-image value – the dictionaries carry the edit-specific signal itself.

Commitment is early; separation is gradual. A pre-registered prediction that D would already localize tightly at every step is refuted, informatively. The edit *content* is present at step 0 (the tuned lens already shows red eyes, a black trunk), but the *differential* at step 0 is a diffuse whole-frame field – the edit prompt perturbs the entire velocity relative to the no-op, not just the edited region – and condenses onto the edit's footprint only gradually (localization of D's top mass at L52 climbs monotonically across steps, e.g. $0.21{\to}0.47{\to}0.67{\to}0.83$ for a cat's-eyes recolor). *Committed* (is the content in the internal image) and *separated* (has the differential found where it lives) are two different quantities, and the value of the no-op control is that it also caught a case where the model *cannot* do nothing – one flat-art scene rendered a transparent-PNG checkerboard for "keep unchanged," which we discard, motivating a control-fidelity gate before any differential number is trusted.

A preview lever. Because the differential carries the edit's footprint, we ask whether it does so early enough to be useful. At step 2 – the third of 20, $\approx$15% of the forward passes – across a 10-case, five-family battery of natural photographs (all passing the control-fidelity gate, $r \in [0.86, 0.99]$), the median IoU between $D_{\text{L52},t2}$'s top mass and the true edit region is 0.33, $6.6\times$ chance, and all 10 cases sharpen from step 0 to step 2 (Figure 7): an object slated for removal lights up as a ghost, a keep-vs-replace background segmentation emerges unsupervised, and a tiny recolor glows at its exact site. (A fixed top-10% ground-truth mask under-scores genuinely tiny edits, padding their region $\sim$$10\times$ with collateral divergence.) Because the no-op control is *independent of the edit prompt*, it can be computed once per source image and cached; previewing where and how large a *candidate* edit will land then costs only the three forward passes to reach step 2 plus one frozen-dictionary decode, not a full 20-step run – a training-free way to triage many candidate prompts against one image before committing to any.

Generalization Across Checkpoints

Every finding so far concerns one checkpoint. We re-ran the depth-lens crossover, the cross-seed probe, and layer-band ablation on two further checkpoints sharing Qwen-Image-Edit-2511's transformer classes: FireRed-Image-Edit-1.1, an

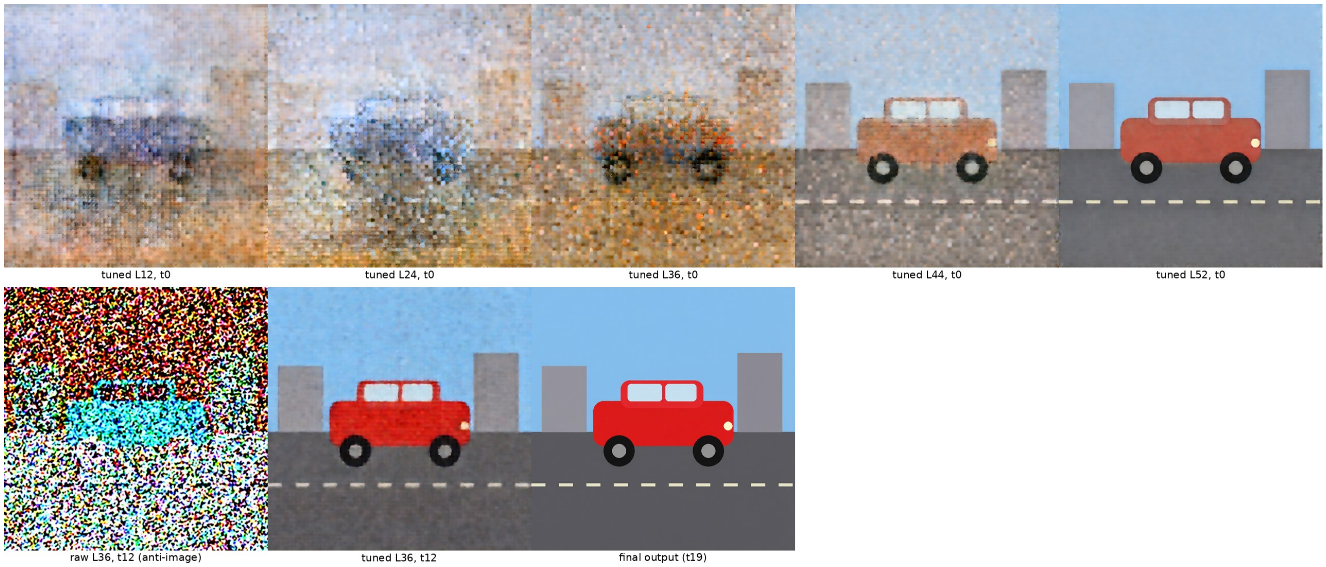

Figure 5: A tuned lens renders the internal image the frozen head cannot read. Top: per-(layer, timestep) ridge translators, fit on *remove/background/addition* edits and evaluated zero-shot on the held-out *car_red* color edit, decode a coherent scene at every depth at t=0, watching the edit enter the internal image layer by layer. Bottom: at t=12 the raw frozen head renders the identical L36 hidden state as a color-inverted ghost – the complement of the true scene – while the tuned lens at the same cell is already coherent.

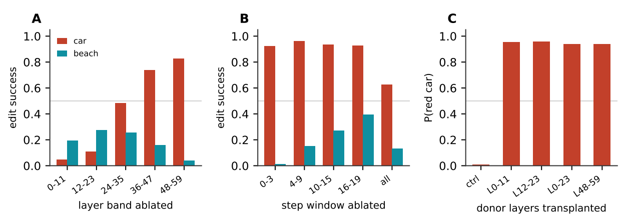

Figure 6: Causal ablation and transplant, *car_red* vs. *beach_background*. **A**: ablating a band of layers shows opposite profiles – car needs its *early* layers, beach needs its *late* ones. **B**: ablating a window of steps shows car tolerates any single window, beach is destroyed by ablating the earliest one. **C**: transplanting a donor *car_red* run's hidden state into a *car_green* recipient, at any of four layer bands, is *sufficient* to flip the output back to red – including the L48–59 band, which panel A shows is *not necessary* for the car edit at all.

independent fine-tune, and Rapid-AIO, a *4-step distilled* merge. **The translation boundary and the early decision replicate.** FireRed's A/C crossover lands at L53–56, Rapid-AIO's at L52–54 – both on Qwen-Image-Edit-2511's L52–54 band; FireRed's probe reads 0.852 at L6 and dips to 0.69 at L59, the same flat-then-dip shape. **The decision direction itself transfers across models**: a probe trained on one model and tested on the *other* reaches 0.84/0.92 accuracy, and the two models' L12 probe directions have cosine similarity 0.833 – the edit lives in nearly the same direction of the shared 3072-d space, not merely the same layer index. The full tuned-lens toolchain also ports to FireRed (deep-layer R^2 0.97–0.999, anti-image reproduced).

One honest divergence, caught by the check doing its job: ablating L48–59 leaves Qwen-Image-Edit-2511's edit intact (0.826) but is catastrophic for FireRed (0.035) – late-stack redundancy is a property of *training*, not the architecture, so the readable$\neq$necessary result reported above, in *Readable, Necessary, Writable*, holds architecture-generally only in its early half.

A Training-Free Lever: Truncating Classifier-Free Guidance

Because the decision is linearly present by layer 6 and translated by layer 53, there is no architectural reason classifier-free guidance needs both of its forward passes at all 20 steps. Running the unconditional pass only for the first two steps and the conditional pass alone thereafter leaves all four edit families visually intact at a **45% reduction in forward passes** (a 20-case battery confirms it, mean 1.77$\times$ speedup). Dropping CFG entirely isolates its real job: recolor, removal, and addition still land from the conditional pass alone, but background swap composites the new scene *around* un-erased source structure ($P(\text{beach})=0.45$ vs. ≥ 0.996) – CFG's non-redundant contribution is resolving *scope*, not committing the edit. The same decision structure survives 7.7$\times$ step distillation (20/20 cases land), so distillation compresses the sampling trajectory but not the decision structure the lens exposes.

Limitations

This tool is exploratory. Every number comes primarily from one model, one or two seeds, and a double-digit count of hand-chosen images; the cross-model checks reported above,

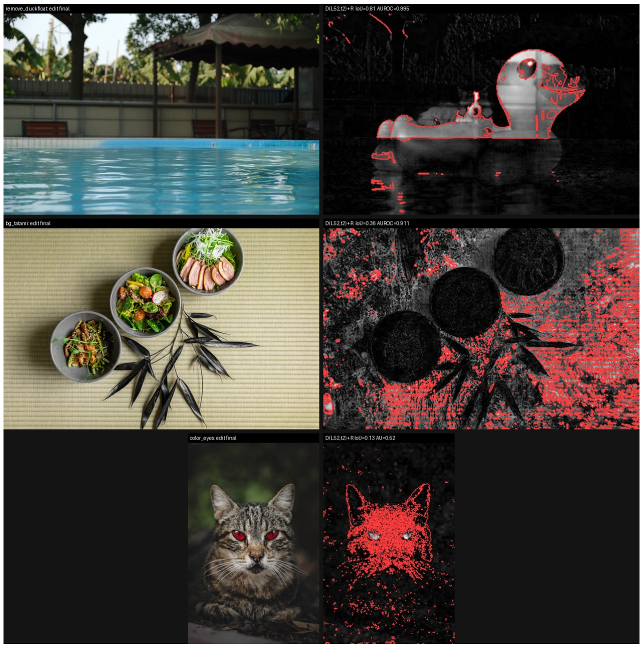

Figure 7: The differential lens as a $\approx$15%-NFE (t=2) preview lever: three of a ten-case battery. Left: each case's final edited output. Right: the differential D at layer 52, step 2, with the true edit region outlined in red – an object slated for removal lights up as a ghost, a keep-vs-replace background segmentation emerges unsupervised, and a tiny recolor glows at its exact site, all at only 15% of the denoising trajectory.

in *Generalization Across Checkpoints*, cover two checkpoints of the same lineage on a subset of measurements, and we make no claim about "intent" or about recovering the network's "true" reasoning. The semantic readout inherits CLIP's failure modes in identifiable ways: pure sampling noise can be mis-scored (a still-noisy decode labeled beach at 0.795), and, in mirror image, an obviously-red recolored car *under*-scores (0.09–0.16) while a *failed* flat patch scores higher (0.70–0.84) – which is why every quantitative claim is paired with a pre-registered visual verdict, and the image wins ties. Four of our own initial misreadings are retained as data: a thumbnail "revision" phenomenon that quantification erased (leaving a cleaner printing-vs-thinking split), a continuous novelty law that did not survive (only a categorical one did), a steering "clean flip" that was a flat patch replacing the car's geometry, and the over-strong conclusion drawn from it – that a direction cannot carry shape – which a silhouette mask refuted. We claim only that, on Qwen-Image-Edit-2511, the readable/necessary/writable dissociations we report are real and repeatable for the seeds and prompts used, because we checked them against causal interventions rather than passive readouts alone.

Conclusion

We built a layer–time and tuned lens for a 60-block DiT image editor and paired every reading with a causal check. The edit is decided early (linearly readable by layer 6), carried as a target-image code for most of the depth, and translated into the velocity code only in a sharp, timestep-invariant band at layer $\approx$53 – a boundary that survives an independent fine-tune and a 4-step distillation, while readability, necessity, and writability dissociate three distinct ways. If one sentence organizes the paper, it is this: *the lens shows the translation, not the decision – reading the decision requires changing basis, and verifying it requires intervention.* Behind it stands the picture we opened with and have now earned clause by clause: an edit is not developed gradually – it is decided early, bound to a spatial carrier, and translated late into the strokes that paint it. A generative editor natively thinks with an internal image before it makes one; the lens turns that thought into something that can be watched, steered, and budgeted for.

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